SOCIAL MEDIA & CONTENT ANALYTICS

Full Insights & Performance Analysis Report

Period: January 2023 - December 2024 | 8 Dashboard Pages | 10 Data Tables

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1. Executive Summary

This report presents a comprehensive analysis of social media and content analytics performance across 8 domains covering 2 full years of data (January 2023 to December 2024). The analysis is built on an enterprise-grade SQLite database containing 8,055 rows across 10 interrelated tables, visualised in Power BI using DirectQuery mode with a custom Midnight Blue and Emerald theme.

A Random Forest machine learning model was trained in Python and integrated into the Power BI dashboard to generate customer churn predictions — creating a live feedback loop between Python ML output and business intelligence visualisation.

Total Reach	Total Revenue	Overall ROAS	Avg Engagement	Total Customers
322.17M	EUR 1.22M	2.46x	6.46%	800

Overall Performance Summary

Domain	Key Metric	Value	Status
Reach & Engagement	Avg Engagement Rate	6.46%	Above target (5%)
Revenue & ROAS	Return on Ad Spend	2.46x	Above target (2.0x)
Audience Growth	Total Followers	266,288	Above target (250K)
Campaign Performance	Total Conversions	151,178	Active: 11 / Completed: 139
Customer Behaviour	Avg Order Value	EUR 811.80	Above target (EUR 780)
Product Performance	Margin Rate	42.16%	Above target (35%)
Geographic Reach	Countries	15	Top: Poland (EUR 104.5K)
Churn & Retention	High Risk Customers	335	Needs attention (target: <200)

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2. Content Performance Analysis

This section analyses 2,000 posts published across 6 social media platforms over 2 years, examining engagement rates, content type effectiveness, and the relationship between sponsored and organic content.

2.1 Key Metrics

Total Posts	Total Impressions	Avg Eng. Rate	Video Views	Sponsored Posts
2,000	570M	6.46%	101M	824

2.2 Platform Performance

Platform	Posts	Total Reach	Avg Eng. Rate	Key Insight
TikTok	~500	227M	~6.4%	Highest reach — 70% of total
Instagram	~600	46M	~6.8%	Best engagement for Reels and Lives
YouTube	~240	31M	~7.1%	Best for Video and Live content
LinkedIn	~180	9M	~6.2%	Best for Articles and Carousels
Facebook	~200	7M	~6.6%	Strong for Video content
X (Twitter)	~280	3M	~5.8%	Lowest reach but decent engagement

2.3 Content Type Effectiveness

The engagement rate heatmap reveals clear patterns across platform and content type combinations:

- Live content delivers the highest average engagement rate overall — particularly strong on TikTok (9.4%) and Instagram (9.2%)
- Reels are consistently high performers across all platforms that support the format — averaging 8.1% engagement
- Video content on YouTube achieves the platform's highest engagement at 8.6% — aligned with audience expectations for the platform
- Static Image and Story formats consistently underperform across all platforms, averaging below 4%
- Poll content shows above-average engagement on TikTok (7.29%) and LinkedIn (6.57%) — two platforms with engaged professional or trend-driven audiences

2.4 Sponsored vs Organic Analysis

Metric	Organic	Sponsored	Winner
Post Volume	1,176 (59.4%)	824 (40.6%)	Organic
Avg Reach per Post	~58K	~112K (+93%)	Sponsored
Avg Engagement Rate	~7.1%	~5.8%	Organic
Avg CTR	~2.6%	~4.6%	Sponsored

> *Key insight: Sponsored posts deliver 93% more reach per post and higher CTR, making them superior for awareness and conversion objectives. However organic posts generate 22% higher engagement rates, making them more effective for community building and authentic audience relationships. The optimal strategy combines both — sponsored for scale, organic for depth.*

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3. Campaign ROI Analysis

150 campaigns were executed across 6 platforms over the 2-year period, with a total media spend of EUR 495,464 generating EUR 1.22M in attributed revenue.

3.1 Key Metrics

Total Campaigns	Total Spend	Revenue	ROAS	Conversions
150	EUR 495K	EUR 1.22M	2.46x	151,178

3.2 ROAS by Platform

Platform	ROAS	vs Target (2.0x)	Status
LinkedIn	2.06x	+0.06x	Above Target
YouTube	1.77x	-0.23x	Below Target
X (Twitter)	1.48x	-0.52x	Below Target
Instagram	1.33x	-0.67x	Below Target
TikTok	1.30x	-0.70x	Below Target
Facebook	1.21x	-0.79x	Below Target

> *LinkedIn is the only platform exceeding the 2.0x ROAS target, suggesting professional audience campaigns deliver higher quality conversions. TikTok drives the most reach but converts at lower rates relative to spend.*

3.3 ROAS by Campaign Objective

Objective	ROAS	Interpretation
Retargeting	2.73x	Best performing — warm audiences convert efficiently
Conversions	1.65x	Direct response performs moderately
Brand Awareness	1.28x	Below break-even — awareness campaigns rarely convert directly
Lead Generation	1.23x	Below target — lead quality or nurturing needs improvement
Engagement	0.67x	Below break-even — engagement spend rarely drives direct revenue

> *Brand Awareness and Engagement campaigns are generating negative ROI. Budget reallocation toward Retargeting and Conversion objectives could significantly improve overall ROAS.*

3.4 Campaign Status

- 139 of 150 campaigns (92.7%) are completed — providing a complete historical dataset for performance analysis
- 11 active campaigns represent ongoing investment with results still accumulating
- Average spend per campaign: EUR 3,303 — relatively small budgets suggesting a diversified testing approach
- Average CPA: EUR 3.28 per conversion — very low, reflecting the high volume of simulated conversions in the dataset

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4. Audience Growth Analysis

Analysis of follower growth across 6 platforms over 24 months, examining acquisition, loss, demographic composition and platform-specific growth trajectories.

4.1 Key Metrics

Total Followers	Net Growth	New Followers	Lost Followers	Avg Monthly Growth
266,288	+81,788	147,710	65,922	2.80%

4.2 Platform Follower Share

Platform	Followers	Share %	Growth %
TikTok	96,865	36.4%	Highest absolute growth
Instagram	70,974	26.7%	Strong consistent growth
Facebook	43,375	16.3%	Steady growth
LinkedIn	25,574	9.6%	Professional audience
YouTube	16,561	6.2%	Moderate growth
X (Twitter)	12,939	4.9%	Slowest growth

4.3 Audience Demographics

Age Band	Share	Insight
18-24	29.37%	Young adult audience — responsive to trends and short-form video
25-34	36.33%	Core demographic — highest purchase intent and engagement
35-44	20.99%	Secondary audience — brand loyal and higher AOV potential
45+	13.31%	Smaller but valuable — typically higher spending power

> *The 25-34 age band representing 36.33% of the audience is the sweet spot for social commerce — this demographic has established spending power, platform familiarity, and the highest engagement rates. Campaign targeting and content strategy should be optimised for this segment first.*

4.4 Follower Retention

The follower churn rate (lost/new) stands at 44.63% — meaning nearly half of newly acquired followers are offset by losses. This is common in social media but represents an opportunity:

- Improving follower retention by 10 percentage points would add approximately 14,771 additional net followers over the same period
- Content consistency and platform-specific posting schedules are the primary levers for reducing follower churn
- Monthly new follower trends are consistently above lost follower trends — net growth is positive and accelerating

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5. Customer Behaviour Analysis

Analysis of 800 customers generating 1,500 orders over the 2-year period, examining acquisition channel quality, segment performance, purchase frequency and lifetime value.

5.1 Key Metrics

Total Customers	Total Orders	Avg Order Value	Customer LTV	Return Rate
800	1,500	EUR 812	EUR 1,522	4.27%

5.2 Revenue by Customer Segment

Segment	Revenue	Share	Avg LTV	Action Required
New	EUR 305,869	25.1%	Lower	Nurture to Potential/Loyal
Potential	EUR 253,293	20.8%	Medium	Activate with targeted campaigns
Loyal	EUR 245,754	20.2%	High	Retain and reward
At Risk	EUR 171,399	14.1%	Medium	Immediate re-engagement
Champion	EUR 133,355	10.9%	Highest	VIP treatment and advocacy
Lost	EUR 108,024	8.9%	Low	Win-back campaigns

> *The New segment generating the highest revenue (25.1%) reflects strong acquisition performance in 2023-2024. However the At Risk segment (EUR 171K) represents recoverable revenue — these customers have demonstrated purchasing behaviour and require targeted re-engagement before transitioning to Lost.*

5.3 Customer Acquisition Channel Quality

Customer LTV varies significantly by acquisition channel, revealing which channels attract the highest-value customers:

- Facebook and TikTok customers generate the highest LTV — suggesting social-native customers have stronger purchase intent
- X (Twitter) and LinkedIn customers show moderate LTV — these channels attract different buyer personas
- Orders per customer averages 1.88 — meaning most customers have made repeat purchases, a positive retention signal
- 607 of 800 customers (75.9%) are subscribed — strong email list for retention campaigns

5.4 Purchase Frequency Analysis

- 42.1% of customers placed only 1 order — the single largest opportunity for revenue growth
- 33.8% placed 2-3 orders — showing initial repeat behaviour that can be reinforced
- 19.9% placed 4-6 orders — loyal repeat buyers worth identifying and rewarding
- 4.2% placed 7+ orders — Champion segment candidates generating disproportionate revenue

> *Converting even 10% of single-order customers to repeat buyers at the average AOV of EUR 812 would generate approximately EUR 33K in incremental revenue — making post-purchase email sequences the highest-ROI retention tactic available.*

6. Product Performance Analysis

Analysis of 50 products across 10 categories generating EUR 1.22M in revenue with a 42.16% margin rate — significantly above the 35% target.

6.1 Key Metrics

Total Products	Total Revenue	Total Margin	Margin Rate	Units Sold
50	EUR 1.22M	EUR 513K	42.16%	4,485

6.2 Category Performance

Category	Revenue Share	Margin %	Status
Books	20.3%	41.1%	Top revenue + healthy margin
Sports	15.x%	~27%	High revenue, below margin target
Beauty	10.59%	~52%	Lower revenue but highest margin
Apparel	10.21%	38.6%	Solid performer
Gaming	9.94%	~41%	Good margin
Nutrition	9.0x%	~35%	At margin target
Software	7.8x%	54.7%	Best margin rate
Home & Living	6.01%	56.1%	Highest margin — low volume
Electronics	5.95%	~22%	Low margin — pricing review needed
Accessories	~5%	53.5%	High margin, low volume

> *Books drives the most revenue (20.3%) with a healthy 41% margin. Software, Home and Living, and Accessories have the highest margin rates (54-56%) but lower volume — these represent premium growth opportunities with targeted marketing spend.*

> *Electronics shows the lowest margin rate at approximately 22% — well below the 35% target. A pricing review or supplier cost renegotiation should be prioritised to bring this category to profitability targets.*

7. Geographic Analysis

15 countries across 4 continents generating EUR 1.22M in revenue, with a relatively even distribution across European markets and emerging growth in South America.

7.1 Key Metrics

Countries	Top Country	Top Revenue	Avg Order Value	Fastest Growing
15	Poland	EUR 104.5K	EUR 812	Ireland

7.2 Country Performance

Country	Customers	Revenue	Avg Order Value	Revenue Share
Poland	57	EUR 104,518	EUR 878	8.58%
Belgium	57	EUR 100,833	EUR 877	8.28%
Netherlands	53	EUR 92,311	EUR 817	7.58%
United Kingdom	61	EUR 91,848	EUR 806	7.54%
Spain	54	EUR 90,978	EUR 892	7.47%
Canada	58	EUR 88,155	EUR 840	7.24%
Australia	48	EUR 87,082	EUR 889	7.15%
Portugal	59	EUR 86,054	EUR 820	7.07%
Brazil	63	EUR 81,964	EUR 719	6.73%

7.3 Geographic Insights

- Revenue is remarkably well distributed across markets — the top country (Poland at 8.58%) represents less than 10% of total revenue, indicating healthy geographic diversification
- Spain leads on average order value (EUR 892) followed by Australia (EUR 889) — high-value markets worth increased marketing investment
- Brazil has the highest customer count (63) but the lowest AOV (EUR 719) — opportunity to upsell and increase basket size through product bundling
- Ireland shows the fastest YoY revenue growth at 24.5% — a market with strong momentum worth accelerating with dedicated campaigns
- United Kingdom has the most customers (61) despite not being the top revenue country — lower AOV suggests price sensitivity or different product mix preferences

> *The geographic distribution suggests a mature multi-market strategy with no single-market dependency. The key growth opportunity lies in increasing AOV in high-volume but lower-AOV markets (Brazil, UK) while accelerating investment in fast-growing markets (Ireland) and high-AOV markets (Spain, Australia).*

8. Retention & Churn Analysis

A Random Forest machine learning model (n_estimators=50, max_depth=4) was trained on customer order behaviour data to predict churn risk. 688 customers with order history were scored, producing a binary classification of High Risk (335) and Low Risk (353).

8.1 Model Performance

Customers Modelled	High Risk	Low Risk	Avg Churn Prob	Revenue at Risk
688	335	353	48.84%	EUR 488K

Model Parameter	Value	Notes
Algorithm	Random Forest Classifier	Ensemble method — robust to overfitting
n_estimators	50	Reduced from 100 for performance
max_depth	4	Constrained to prevent overfitting on small dataset
class_weight	Balanced	Accounts for class imbalance
ROC-AUC Score	0.87	Strong discriminative performance
Train/Test Split	80/20	Stratified by churn label
Churn Definition	No order in last 90 days	As of 2024-12-31

8.2 Key Features Driving Churn

- Days Since Last Order — strongest predictor (high weight in feature importance)
- Total Orders — customers with more orders are significantly less likely to churn
- Total Revenue — higher LTV customers show lower churn probability
- Average Order Value — lower AOV correlates with higher churn risk
- Customer Segment — At Risk and Lost segments dominate the High Risk prediction band

8.3 Churn by Acquisition Channel

Channel	Churn Rate	Risk Level	Recommended Action
Email	56.3%	High	Review email campaign quality and frequency
Facebook	53.8%	High	Improve post-acquisition engagement for FB customers
TikTok	50.4%	High	TikTok customers may be impulse buyers — improve retention
LinkedIn	42.2%	Medium	Best performing channel for retention
Instagram	46.8%	Medium	Moderate retention — improve with personalisation

> *Email channel showing 56.3% churn rate is counterintuitive — email is typically a strong retention channel. This suggests email acquisition may be attracting lower-intent customers, or post-purchase email sequences are underperforming. A full email journey audit is recommended.*

8.4 Revenue at Risk

- 335 high-risk customers represent EUR 488,400 in historical revenue — an average of EUR 1,458 per at-risk customer
- If 30% of high-risk customers are successfully retained through targeted campaigns, this represents EUR 146,520 in protected revenue
- The average days since last order for high-risk customers is 220.9 days — these customers have been inactive for over 7 months
- Low-risk customers average just 36.4 days since last order — demonstrating clear behavioural separation between risk bands

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9. Recommendations & Next Steps

9.1 Immediate Priority Actions (0-30 Days)

- Launch win-back campaign for 335 high-risk customers — personalised email or social retargeting using churn probability scores
- Reallocate campaign budget from Engagement and Brand Awareness objectives toward Retargeting — projected ROAS improvement from current 2.46x to 3.0x+
- Review Electronics pricing strategy — 22% margin rate is below target and requires immediate action
- Activate post-purchase email sequence for single-order customers — 336 customers have placed exactly one order and represent the highest revenue growth opportunity

9.2 Short-Term Improvements (1-3 Months)

- Increase Live content production for TikTok and Instagram — Live format achieves 9.2-9.4% engagement rate vs 6.46% average
- Develop Spain and Australia market strategies — both show above-average AOV (EUR 892 and EUR 889) with room for volume growth
- Implement LinkedIn-focused B2B campaigns — LinkedIn is the only platform exceeding ROAS target (2.06x) and has the lowest churn rate (42.2%)
- Introduce loyalty programme for Champion and Loyal segments — these 245 customers generate 31.1% of revenue from 30.6% of the customer base

9.3 Strategic Initiatives (3-12 Months)

- Expand into underserved age demographics — 45+ represents 13.31% of audience but typically higher spending power; targeted content could yield premium returns
- Build influencer partnership programme across Nano and Micro tiers — these deliver the best engagement rate per cost
- Develop Ireland market acceleration plan — 24.5% YoY growth makes this the most promising expansion market
- Refresh churn model quarterly with new order data — the 90-day churn definition and model parameters should be validated as the dataset grows

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10. Data Notes & Methodology

10.1 Dataset Overview

Table	Rows	Key Columns
Posts	2,000	Platform, ContentType, PostDate, Reach, Impressions
Engagement	2,000	EngagementRate, CTR, VirScore, VideoViews
Campaigns	150	Platform, Objective, Budget, Status
Campaign_Performance	600	Spend, Revenue, Conversions, ROAS, CTR
Customers	800	Segment, AcqChannel, Country, JoinDate, Age
Orders	1,500	Revenue, Margin, Quantity, Platform, OrderDate
Products	50	Category, Price, MarginPct, IsActive
Audience	144	Platform, TotalFollowers, NewFollowers, LostFollowers
Influencers	80	Tier, Followers, EngagementRate, ROAS, Fee
Churn_Predictions	688	ChurnProbability, ChurnRisk, DaysSinceLastOrder

10.2 Known Data Issues

- Campaign_Performance[Revenue] is inflated synthetic data (EUR 24.36M) — all revenue metrics use Orders[Revenue] (EUR 1.22M) as the correct figure
- Avg CTR card on Content page initially displayed as 570M due to column format — the correct value is 2.60%
- Reach YoY % shows +106.9% due to a Calendar relationship issue — the actual growth is approximately +18% based on platform reach comparison
- Total Products card shows 50 items in the product table — the dashboard figure of 800 on one view was a display error (segment count)
- No Medium Risk customers exist in the churn output — all 688 customers fell in either Low (0-30% probability) or High (60-100%) bands, with the 30-60% middle band being empty

10.3 Methodology

- All data is simulated for portfolio purposes using Python with Faker and controlled random distributions
- Revenue is always calculated from Orders[Revenue] not Campaign_Performance[Revenue] to avoid inflation
- Total Followers uses MAX per platform (not SUM) to avoid summing all monthly rows
- YoY calculations use direct year filters (FILTER ALL Calendar Year=2023) rather than SAMEPERIODLASTYEAR due to many-to-many Calendar relationships
- Churn definition: no order placed in the 90 days preceding 2024-12-31

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11. Conclusion

This Social Media and Content Analytics dashboard represents a full-stack data project combining Python data engineering, SQL analytical scripting, machine learning, and Power BI visualisation — all grounded in genuine commercial and marketing analytics principles.

The most significant findings are the strong ROAS performance (2.46x against a 2.0x target), the revenue concentration risk in the New customer segment (25.1% of revenue from customers with no established loyalty), and the churn risk profile showing 335 customers (EUR 488K revenue) requiring immediate retention intervention.

The integration of a Python Random Forest churn model directly into Power BI creates a live intelligence layer that transforms the dashboard from a reporting tool into a decision-support system — the defining feature of this project from a technical portfolio perspective.

-- End of Report --